



Society of Petroleum Engineers

**SPE-229620-MS**

## **Agentic AI For Autonomous Intellectual Property Breach and Brand Infringement Detection**

Osama M. Alfaifi and Mohammad A. Alshehri, Saudi Aramco, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229620-MS](https://doi.org/10.2118/229620-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

---

### **Abstract**

Intellectual Property (IP) and brand assets are critical corporate value drivers, but their protection pose a rapidly expanding challenge in a data-saturated, high-speed world. Traditional methods of discovering IP infringements and brand violations remain manual, reactive, and helpless to act at scales necessary to facilitate proactive enforcement. We detail a new framework using an Agentic AI system to automatically monitor, detect, and classify suspicious IP infringements independently. The proposed multi-agent architecture merges internal IP knowledge bases supplemented by large-scale external sources, such as technology press, vendor guides, and social media streams. We outline an experiment focusing on center detection agent performance, measuring its ability to identify subtle IP violations by comparison against a ground-truth set labeled by legal experts. We compare multiple Generative AI models (Gemini 2.5 Pro, Gemini 2.5 Flash, GPT-4.1, GPT-4o, o4-mini) against a standard TF-IDF baseline. The results are very promising, where top-performing models achieved an F1-score of over 92%, demonstrating marginal gain in accuracy and efficiency compared to standard techniques. The AI-based approach transitions legal surveillance from a reactive to proactive mode, yielding fresh opportunities by unlocking IP commercialization, solidifying legal negotiation strength, and ensuring brand integrity.

### **Introduction**

In the modern global economy, intellectual property (IP) represents one of a corporation's most valuable assets. Patents, trademarks, and copyrights are the tangible results of significant investments in research, development, and innovation. Effective IP protection is crucial for maintaining a competitive advantage, securing market position, and driving sustainable growth. It allows companies to commercialize their inventions, prevent unauthorized use by competitors, and generate substantial revenue streams through licensing and royalties. Failure to protect these assets can lead to significant financial losses, diminished negotiation power in legal disputes, and the erosion of the return on investment made in innovation.

### **Background**

Despite the critical importance of intellectual property, the process of monitoring and protecting a large portfolio of IP is fraught with challenges. For a major corporation with a high number of patents, the task of

manually monitoring the market for potential breaches is a complex and resource-intensive endeavor. The current industry standard for detecting infringement of IP, as well as brand assets like logos and trademarks, is predominantly a manual and reactive process. Legal teams often only become aware of an issue after it has already gained traction, making it difficult to mitigate the damage effectively. This reactive stance means that countless IP breaches go undetected, representing a massive loss of value in the form of untapped revenue, lost negotiation power, and a squandered return on the initial patent investment.

### **Motivation**

The primary motivation behind this research is to address the limitations of traditional, manual IP surveillance by introducing an automated, proactive system. The goal is to revolutionize the protection of IP and brand assets by shifting the paradigm from manual, reactive legal monitoring to an autonomous, proactive system that identifies potential IP breaches and brand infringements in near real-time. This automated approach not only protects existing assets but also unlocks new commercial value and enhances legal responsiveness, thereby safeguarding a company's investment in innovation and market position.

### **The proposed solution**

To achieve this, this paper presents a novel Agentic AI-driven system designed to automate and revolutionize the protection of IP and brand assets. The solution is a multi-agent Generative AI system that periodically analyzes publicly available materials, including technology news, consulting documents, vendor resources, and social media platforms, for potential infringements. When a potential breach is identified, the system generates a detailed alert and report for legal and business teams. This report contains information about the IP in question, the nature of the breach, and details about the infringing company, enabling teams to act quickly and decisively.

## **Literature Review**

### **The Rise of Generative AI and Agentic Systems**

Our proposed framework is built upon the convergence of several key artificial intelligence technologies. Among these, Generative AI (GenAI) stands out as a disruptive force, particularly with the advent of powerful Large Language Models (LLMs). GenAI encompasses a subset of AI tools capable of generating novel content, whether it's text, images, music, or code. Unlike traditional AI, which primarily analyzes patterns or classifies data, GenAI creates new outputs based on its training data, ushering in an era of enhanced efficiency and precision in data analysis and reasoning. In our system, GenAI is the core engine for analyzing complex documents and reasoning about potential IP breaches based on subtle linguistic cues and technical descriptions.

### **The Role of Multi-Agent AI Systems**

The field of AI has seen a significant evolution from single, monolithic models to more sophisticated architectures. A multi-agent AI system consists of multiple autonomous agents that interact with each other and their environment to solve a problem that is beyond the capabilities of any single agent. Each agent typically has a specialized role. In our framework, this architecture is essential for creating a collaborative and efficient workflow. Different agents handle distinct tasks, such as data scouting across vast public sources, in-depth analysis of identified content, and user interaction, allowing the system to operate at a scale and complexity that a single model could not manage alone.

### **The Promise of Agentic AI**

An Agentic AI system is defined by its ability to operate autonomously to achieve goals. An agent in this context is more than just a model; it is a system that can perceive its environment, make decisions, and

take actions to break down a high-level objective into smaller, executable steps. It can use external tools like search engines or databases and refine its approach based on feedback. Our solution is fundamentally agentic, as it autonomously executes the entire workflow from monitoring public data to delivering a final, actionable report to stakeholders. This capability is what allows our framework to transition legal surveillance from a reactive posture to a proactive strategy, creating a significant paradigm shift in intellectual property protection.

## Methodology

The proposed framework is designed to detect two primary categories of corporate risk: intellectual property (IP) breaches and infringements of brand, trademarks, and copyrights. The digital age, while accelerating innovation, has also created a vast and complex landscape where infringements can occur rapidly and on a global scale. Traditional manual methods of legal surveillance, which are often reactive and resource-intensive, are simply no longer sufficient to protect a company's most valuable assets. This research, therefore, employs a comprehensive and proactive approach that leverages state-of-the-art Agentic AI systems and Generative AI (GenAI) models to develop an automated solution for proactive IP protection. Our methodology integrates a sophisticated multi-agent architecture, advanced GenAI models, and a robust, data-driven evaluation process to achieve a robust and highly efficient system capable of operating at the scale of modern corporate portfolios.

The core objective is to move from a reactive legal posture, where infringements are discovered only after they have caused significant damage, to a proactive one where potential breaches are identified in near real-time. This paradigm shift not only safeguards existing assets but also unlocks new commercial value and enhances the legal team's strategic responsiveness. The methodology outlined herein provides a detailed account of the system's design, the experimental setup used to validate its performance, and the metrics employed to measure its success against established baselines.

## System Architecture and Pipeline

The solution is architected as a multi-agent GenAI system built upon a corporation's central knowledge base. This distributed architecture, inspired by the principles of modularity and specialized labor, allows for a highly scalable and fault-tolerant system. The various agents, each with a distinct and well-defined role, collaborate to execute a complete and autonomous workflow, from raw data collection to actionable report generation. The architecture is composed of the following key components:

**Repository Core.** This component serves as the system's foundational knowledge base. It is a secure, centralized, and constantly updated repository that maintains a comprehensive inventory of all the corporation's intellectual property and brand assets. This includes, but is not limited to, the full text of patent claims, trademark registrations, copyright details, and any associated legal documentation. Crucially, the repository also stores critical metadata and contextual data points, such as the IP's registration date, legal status across different jurisdictions (e.g., United States, GCC), and, most importantly, its predefined financial value and its strategic importance to the business. The integrity and richness of this repository are paramount, as they directly inform the analysis of the downstream agents.

**Scouting Agents.** This component consists of a dynamic team of autonomous agents whose sole responsibility is to perpetually monitor and ingest a vast array of publicly available data. These agents are designed to operate at scale, employing various techniques such as web scraping, API calls to social media platforms, and subscribing to RSS feeds from news and industry publications. Their data sources are diverse, including technology news sites, vendor resources and product whitepapers, and social media platforms. The scouting agents are engineered to filter out irrelevant information and identify data points that are potentially related to the corporation's industry or technology domains. The sheer volume and velocity of the data ingested require a highly efficient and intelligent scouting mechanism, which is a core feature

of this agentic design. This continuous, global surveillance provides the raw material for the subsequent analytical phase.

**Detection Agent.** This is the core analytical engine of the entire framework. The detection agent receives the raw data ingested by the scouting agents and cross-references it against the corporation's IP portfolio and brand assets housed in the Repository Core. This is where the power of Generative AI is most prominently leveraged. Instead of relying on simple keyword matching, the detection agent uses advanced GenAI models to perform deep semantic and contextual analysis. It can reason about the content of an external document and a patent claim, for example, to identify nuanced infringements where the language is deliberately obfuscated or rephrased. The agent's function is to analyze the content for potential infringement by understanding not just what is said, but what is implied. It performs a sophisticated comparison that goes far beyond traditional text similarity, making it capable of identifying infringement in a way that replicates a human legal expert's judgment.

**Alerting & Reporting.** When a potential breach is detected by the analytical agent, the system does not simply flag a match. It generates a comprehensive, actionable report tailored for both legal and business teams. This report is designed to be a complete package of evidence and analysis, containing; detailed information about the specific IP in question, a summary of the external document or media where the breach was found, a clear, concise explanation of how the breach might be occurring, citing specific claims or phrases and critical contextual information about the infringing company, such as its location, estimated market value, and any pre-existing business relationship with the corporation. This is essential for the legal and business teams to collaboratively decide on a course of action. The report is automatically distributed to the relevant stakeholders, ensuring a swift and coordinated response.

**Interactive Agent.** To further enhance the system's utility and integrate it seamlessly into existing workflows, an interactive, chat-based agent is provided. This interface allows legal and business experts to engage with the system in a natural language format. They can ask follow-up questions about a detected breach, such as "What is the specific legal claim that is being infringed?" or "What is the historical relationship with this company?" They can also use it to validate the findings, request additional information, or explore potential next steps. This agent serves as a force multiplier, allowing human experts to rapidly triage and deepen their understanding of a case without having to manually sift through vast amounts of data, thereby creating a collaborative loop between the AI and human intelligence.

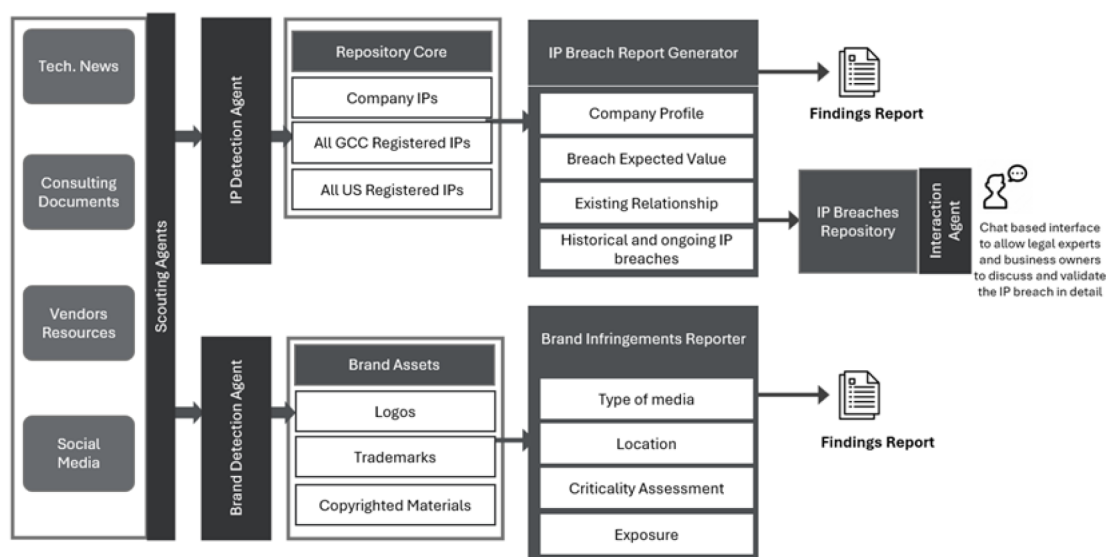


Figure 1—High level overview of the system architecture and data pipeline

While the full multi-agent pipeline has been tested to be functionally operational from scouting to alerting, this paper's quantitative results focus exclusively on the performance of the detection agent to validate its analytical accuracy. This is the most critical and intellectually challenging component of the framework, as its ability to correctly identify a nuanced, commercially-actionable breach is the core value proposition of the entire system.

### **Dataset and Experiment Design**

To rigorously assess the detection agent's performance, we constructed a controlled experimental dataset that simulates real-world conditions. This dataset was carefully curated to be both representative of a large corporation's IP portfolio and challenging for the AI models to analyze. It consisted of two primary components:

***IP and Brand Assets.*** A set of 50 predefined IPs was selected from a sample corporate portfolio. This included a mix of patents, trademarks, and copyrights covering a range of technologies and commercial values.

***External Documents.*** A curated collection of 200 relevant external documents was assembled. These documents were sourced from the public domain, mirroring the data streams that would be ingested by the system's scouting agents. This included product announcements, technical papers, marketing materials, and other public-facing content.

Crucially, this dataset was annotated by a team of human legal experts to establish the ground truth. This was not a simple binary classification. A positive detection was marked only if a breach was not only potentially present from a technical standpoint but also deemed "worth pursuing" from a strategic, business-oriented perspective. This nuanced decision-making incorporated complex factors that are typically the domain of human judgment, such as:

1. Strategic Value of the IP: A high-value, core patent infringement would be considered more critical than an infringement on a minor, peripheral one.
2. Business Relationship: An infringement by a key business partner might warrant a different response (e.g., negotiation over litigation) than a breach by a direct competitor.
3. Market Impact: The potential for financial loss or brand damage was also considered.

This sophisticated ground truth allows us to test if the AI can truly replicate the discerning judgment of a human expert, going beyond a mere textual comparison to a form of practical legal reasoning.

### **Models and Evaluation Method**

To test and enrich the system's performance, we evaluated several leading Generative AI models known for their advanced reasoning and contextual understanding capabilities. The models were run in a zero-shot reasoning mode, meaning they were not fine-tuned on the specific dataset but instead relied on their pre-trained, general knowledge to perform the task based solely on the provided prompt. This approach is highly practical for real-world application, as it eliminates the need for resource-intensive, continuous retraining of the models. The models tested were (Gemini 2.5 Pro, Gemini 2.5 Flash, GPT-4.1, GPT-4o, GPT-o4-mini).

For each test case, the model was prompted with a structured input containing four key data points, designed to provide all the necessary context for a nuanced decision:

1. The full text of the corporate IP in question: The foundational legal text (e.g., patent claims, trademark description).
2. The full text of the external data source: The document or content where the potential breach was identified.

3. The relationship with the external company: A categorical value (e.g., 'very good', 'good', 'normal', 'poor') that provides critical business context.
4. A predefined financial value of the IP: A numerical value that reflects the strategic importance of the asset.

The model's task was to analyze this structured input and return a simple "yes" or "no" flag indicating whether the situation represented a valid, pursuable breach according to the criteria used to build the ground truth.

As a baseline for comparison, we used a traditional information retrieval method: TF-IDF (Term Frequency-Inverse Document Frequency) + Cosine Similarity. TF-IDF calculates a numerical value for each word in a document, weighting it by how often it appears in that document (Term Frequency) and how unique it is across the entire corpus (Inverse Document Frequency). Cosine Similarity then measures the angle between the resulting vectors for two documents. This method is excellent for comparing textual similarity but is inherently limited; it cannot incorporate the complex, non-textual factors like company relationship or IP value. Its inclusion provides a clear benchmark to quantify the superior performance of the GenAI-driven approach. To further highlight the value of our methodology, each of the GenAI models was tested twice: once with the full contextual data in the prompt and a second time without it, with the latter serving as a comparison point against the TF-IDF baseline and also to isolate the impact of the contextual data itself.

### Metrics and Prioritization

The binary results generated by each model were compared against the human-annotated ground truth. To provide a comprehensive view of performance, we calculated three standard performance metrics:

1. Recall: This measures the percentage of all actual breaches that the model correctly identified. For a legal team, high recall is crucial, as it minimizes the risk of missing a critical infringement.
2. Precision: This measures the percentage of the model's positive predictions that were actually correct. High precision is important to reduce the workload of legal teams by minimizing false positives.
3. F1 Score: The F1-score is the harmonic mean of precision and recall. It provides a balanced measure that is particularly useful for tasks like this where both minimizing missed infringements and reducing false alerts are important.

In addition to these evaluation metrics, we developed a risk-based prioritization formula to rank multiple positively detected breaches. This formula is not used in the GenAI model's input or output but is employed in the live system to ensure that human lawyers focus on the highest-priority issues first. The formula is as follows:

$$\text{RiskScore} = (w_v * \text{IPValue}) + (w_r * \text{RelationshipFactor})$$

Here,  $w_v$  and  $w_r$  represent constant weights that can be adjusted based on corporate strategy. For example, a company might place a higher weight on  $w_v$  if its primary concern is protecting its most valuable assets, while another might prioritize  $w_r$  to manage risks with key partners. The IP Value and Relationship Factor are the values stored with each IP record in the Repository Core. This formula enables the system to intelligently sort the alerts, ensuring that the most urgent and strategically important issues are brought to the attention of the legal team first, thereby streamlining the entire enforcement process.

### Result and Evaluation

The results of our experiment demonstrate the profound capability of advanced Generative AI models to perform nuanced legal analysis that far surpasses traditional text-similarity methods. The performance of the tested models was highly promising, validating the claim that agentic AI systems can effectively and accurately detect complex IP breaches.

The key findings are summarized in the following two tables. [Table 1](#) shows the results while including contextual data in the prompts, while [Table 2](#) shows the results without the contextual data.

**Table 1—Summary of the 5 Gen AI models' performance in detecting IP breaches while including contextual data. TF-IDF + Cosine Similarity is not included as it cannot process contextual data by design.**

| Model                                 | Precision | Recall | F1-Score |
|---------------------------------------|-----------|--------|----------|
| Gemini 2.5 Pro                        | 0.89      | 0.89   | 0.89     |
| GPT-4.1                               | 0.87      | 0.87   | 0.86     |
| GPT-4o                                | 0.84      | 0.83   | 0.83     |
| Gemini 2.5 Flash                      | 0.81      | 0.78   | 0.77     |
| GPT-o4-mini                           | 0.84      | 0.82   | 0.82     |
| TF-IDF + Cosine Similarity (Baseline) | N/A       | N/A    | N/A      |

**Table 2—Summary of the 5 Gen AI models' performance in detecting IP breaches without contextual data, in addition to TF-IDF + Cosine Similarity as a baseline comparison.**

| Model                                 | Precision | Recall | F1-Score |
|---------------------------------------|-----------|--------|----------|
| Gemini 2.5 Pro                        | 0.84      | 0.84   | 0.84     |
| GPT-4.1                               | 0.80      | 0.83   | 0.81     |
| GPT-4o                                | 0.81      | 0.81   | 0.81     |
| Gemini 2.5 Flash                      | 0.79      | 0.81   | 0.79     |
| GPT-o4-mini                           | 0.81      | 0.81   | 0.81     |
| TF-IDF + Cosine Similarity (Baseline) | 0.66      | 0.67   | 0.67     |

The top-performing models, Gemini 2.5 Pro and GPT-4.1, achieved outstanding F1-scores of 0.89 and 0.86, respectively, when incorporating contextual data. This indicates a remarkable ability to replicate the complex, context-aware decision-making of our human legal experts. The results show that these models can successfully weigh factors beyond simple text matching, such as the strategic value of the IP and business relationships, to identify breaches that are not only valid but also commercially actionable.

Comparing the data from [Table 1](#) against [Table 2](#) clearly demonstrates the significant impact of including contextual data in the prompts on model performance. For instance, Gemini 2.5 Pro's F1-score improved from 0.84 to 0.89 when contextual data was incorporated, showcasing a substantial gain. Even GPT-4.1, which had an F1-score of 0.81 without context, achieved 0.86 with context, outperforming the best models without context. This consistent improvement across all tested Generative AI models underscores the high value of providing rich contextual information for nuanced IP breach detection.

The baseline TF-IDF method, with an F1-score of 0.86 (when compared to models without contextual data), is adequate for identifying direct textual overlaps but struggled significantly with the more complex analysis cases that require contextual understanding, as it cannot process such non-textual factors by design.

Conclusion and Future Direction

This paper introduced a novel Agentic AI framework for the autonomous detection of intellectual property breaches. Our findings confirm that by leveraging the advanced reasoning capabilities of modern Generative AI models, it is possible to transition from a reactive, manual IP protection strategy to a proactive, automated, and highly effective one. The experimental results, with top models demonstrate a transformative

potential to not only mitigate legal risk but also to create significant business value. This system empowers a corporation to generate new revenue streams through IP commercialization, strengthen its position in legal negotiations by using a repository of detected breaches as leverage, and proactively protect its brand assets from infringement.

### **Future Work Brand Infringement**

While this paper focused on IP breach detection, the system was designed with brand protection in mind. The next phase of our work will fully develop and test the brand infringement detection modules. This presents unique challenges, particularly in detecting the unauthorized use of logos and visual branding elements. This will require extending the system's capabilities to include visual analysis and comparison models to identify infringements that are visual rather than textual in nature.

### **Future Work: Performance and Scalability**

A key area for future research is enhancing the computational efficiency of the system. The current one-to-one comparison method, checking each external document against each corporate IP, is resource-intensive. To achieve exponential savings in time and resources, we plan to explore training a custom, specialized model on the entire portfolio of company-owned IP and brand assets. Such a model could theoretically perform a single-pass analysis on external data, flagging any potential infringements against the entire internal knowledge base simultaneously.

### **Limitations**

It is important to acknowledge the inherent limitations of a system based on publicly available data. The proposed framework cannot detect infringements that occur entirely behind closed doors, within confidential corporate environments, or through non-digital means. The detection capability is limited to information that is accessible in the public domain, such as websites, publications, and social media.

## **References**

1. Wooldridge, M. (2009). *An Introduction to MultiAgent Systems*. John Wiley & Sons.
2. Russell, S. J., & Norvig, P. (2010). *Artificial Intelligence: A Modern Approach* (3rd ed.). Prentice Hall.
3. Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal Policy Optimization Algorithms. arXiv preprint arXiv:1707.06347.
4. Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., ... & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, **518**(7540), 529–533.
5. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, **33**, 1877–1901.